

Chapter 01: Introduction to Differential Equations & Initial-Value Problems

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Course: Applied Ordinary Differential Equations (ENGR 213) | Textbook: Dennis G. Zill (7th Ed., Chapter 1)

1. Executive Summary & First-Principles Philosophy

In engineering, physical laws rarely tell us directly what a quantity is; instead, they tell us **how that quantity changes**. Newton's second law ($F = ma = m \frac{d^2x}{dt^2}$), Fourier's law of heat conduction, and Kirchhoff's circuit laws are all equations relating an unknown function to its derivatives. A **Differential Equation (DE)** is simply an equation containing the derivatives of one or more unknown functions.

Solving a differential equation means working backwards: given the rates of change and geometrical constraints, discover the underlying function $y(x)$ that generated them.

2. Core Mechanics & Mathematical Engine

A. Classification by Type, Order, and Linearity

Every differential equation in ENGR 213 must be classified across three fundamental axes:

1. Type:

- **Ordinary Differential Equation (ODE)**: Contains only ordinary derivatives of one or more unknown functions with respect to a **single independent variable** (e.g., $\frac{dy}{dx} + 5y = e^x$).
- **Partial Differential Equation (PDE)**: Involves partial derivatives of an unknown function with respect to **two or more independent variables** (e.g., $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$).

2. Order:

- The order of a differential equation is the order of the **highest derivative** present in the equation.
- *Example*

: $\frac{d^2y}{dx^2} + 5 \left(\frac{dy}{dx} \right)^3 - 4y = 0$ is **Second-Order** (because the highest derivative is y'' , even though the first derivative is cubed).

3. Linearity:

- An n -th order ODE is **linear** if it can be written in the form:

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- **Two Essential Non-Linearity Traps:**

1. The dependent variable y and all its derivatives $y', y'', \dots$ appear strictly to the **first power** (no $y^2, (y')^3, \sqrt{y}$).
2. No nonlinear functions of the dependent variable exist (no $\sin(y), e^y, \ln(y)$) and no cross-products (no $y \cdot y'$).

B. Initial-Value Problems & The Picard-Lindelöf Theorem

An **Initial-Value Problem (IVP)** seeks a solution $y(x)$ to an ODE subject to side conditions specified at a **single point** x_0 :

$$y' = f(x, y), \quad y(x_0) = y_0$$

Theorem 1.2.1 (Existence and Uniqueness of a Unique Solution): Let R be a rectangular region in the xy -plane defined by $a \leq x \leq b, c \leq y \leq d$ that contains the point (x_0, y_0) in its interior. If both:

1. $f(x, y)$ is **continuous** on R , and
2. $\frac{\partial f}{\partial y}$ is **continuous** on R ,

then there exists some open interval $I_0 : (x_0 - h, x_0 + h)$ contained in $[a, b]$, and a **unique function** $y(x)$ defined on I_0 that solves the IVP.

3. Curriculum-Grounded Visual Reference

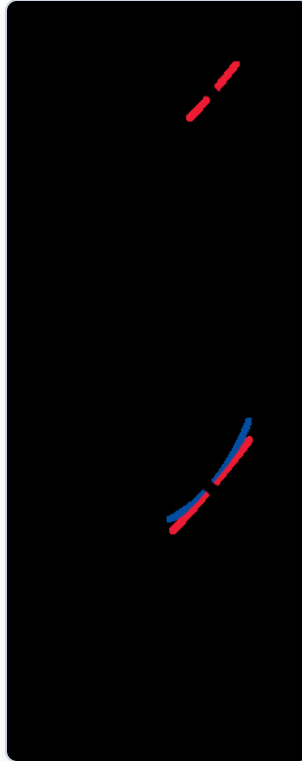


Figure 1.1: Lineal element indicating slope at point (x, y) — from Zill 7th Ed. Chapter 2 (Fig. 2.1.1).

Deep Pedagogical Breakdown:

1. At any coordinate point (x_0, y_0) , the differential equation $\frac{dy}{dx} = f(x, y)$ assigns a numerical slope $m = f(x_0, y_0)$.
2. We represent this geometrically as a tiny line segment (lineal element) passing through (x_0, y_0) with slope m .
3. Any valid solution curve $y(x)$ must pass through the field such that its tangent line at every point coincides exactly with the direction of the lineal element.

4. Fully Worked Exam Archetype

Problem: Determine whether Theorem 1.2.1 guarantees a unique solution for the IVP:

$$\frac{dy}{dx} = y^{1/3}, \quad y(0) = 0$$

Step-by-Step Solution:

- **Step 1: Check continuity of $f(x, y)$:**

$$f(x, y) = y^{1/3}$$

$f(x, y)$ is continuous for all $(x, y) \in \mathbb{R}^2$, including at $(0, 0)$. Existence of a solution is guaranteed!

- **Step 2: Check continuity of $\frac{\partial f}{\partial y}$:**

$$\frac{\partial f}{\partial y} = \frac{1}{3}y^{-2/3} = \frac{1}{3y^{2/3}}$$

At the initial condition point $(0, 0)$, $y = 0$, so $\frac{\partial f}{\partial y}$ is **undefined** (division by zero) and discontinuous!

- **Step 3: Conclusion & Physical Consequence:** Uniqueness is **NOT guaranteed**. Indeed, separating variables gives:

$$\int y^{-1/3} dy = \int dx \implies \frac{3}{2}y^{2/3} = x + C$$

With $y(0) = 0$, we find $C = 0 \implies y(x) = \left(\frac{2}{3}x\right)^{3/2}$. Notice that $y(x) = 0$ is also an obvious valid solution satisfying $y(0) = 0$. Thus, there are infinitely many solutions passing through $(0, 0)$!

5. Rapid Exam Traps & Red Flags

- **⚠ Trap 1: Confusing Order and Degree:** $\left(\frac{dy}{dx}\right)^4 + y = 0$ is **First-Order**, degree 4. The order is determined solely by the number of tick marks / derivative order, not exponents.
- **⚠ Trap 2: Overlooking Explicit vs Implicit Solutions:** An equation $G(x, y) = 0$ is an implicit solution if it defines one or more explicit solutions without explicitly isolating y . On exams, always check whether the professor asks for an explicit solution ($y = \dots$) or accepts an implicit relation.